ASSIGNMENT-1
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1. Relationship between Resistivity and Conductivity

Resistivity and conductivity are opposite quantities. A material with high conductivity has low resistivity and vice versa.

2. Node, Bonding and Anti-bonding Orbital
Node

A node is a region where the probability of finding an electron is zero.

Bonding Orbital

It is formed by constructive overlap of atomic orbitals and makes the molecule stable.

Anti-bonding Orbital

It is formed by destructive overlap of atomic orbitals and makes the molecule unstable.

3. Pauli's Exclusion Principle

No two electrons in an atom can have the same set of four quantum numbers. An orbital can contain a maximum of two electrons with opposite spins.

4. Difference between Metal, Semiconductor and Insulator

Metal

Metals have very high conductivity because their energy bands overlap.

Semiconductor

Semiconductors have moderate

conductivity and a small energy gap.

Insulator

Insulators have very low conductivity due to a large energy gap.

5. Fermi Energy

Fermi energy is the highest occupied energy level of electrons at absolute zero temperature.

6. Classification of Semiconductors
Intrinsic Semiconductor

Pure semiconductor without impurities.

Extrinsic Semiconductor

Semiconductor doped with impurities.

n-type Semiconductor

Formed by adding pentavalent impurities.

Electrons are majority carriers.

p-type Semiconductor

Formed by adding trivalent impurities.
Holes are majority carriers.

7. Direct and Indirect Band Gap Semiconductor
Direct Band Gap Semiconductor

Electrons recombine directly and emit light efficiently.

Indirect Band Gap Semiconductor

Electron transition requires additional energy transfer, so light emission is weak.

8. Properties of Semiconductor
Moderate conductivity
Small energy gap
Conductivity increases with temperature
Current flows through electrons and

holes
Conductitivity can be controlled by doping
9. Barrier Potential and Depletion Width

Barrier potential is the potential difference developed across a p-n junction.
Depletion width is the region around the junction where free charge carriers are absent.

10. Mass Action Law

At thermal equilibrium, the product of electron and hole concentrations remains constant.

11. Effective Mass

Effective mass is the apparent mass of an electron or hole inside a crystal when an external force is applied.

12. Drift, Generation and

Recombination
Drift

Movement of charge carriers due to an electric field.

Generation

Creation of electron-hole pairs due to thermal energy.

Recombination

Process in which electrons combine with holes.

13. Position of Fermi Level
Intrinsic Semiconductor

Fermi level lies near the center of the band gap.

n-type Semiconductor

Fermi level shifts towards the conduction band.

p-type Semiconductor

Fermi level shifts towards the valence band.

14. Diffusion and Einstein Relation
Diffusion

Movement of charge carriers from higher concentration to lower concentration.

Einstein Relation

It relates diffusion coefficient and mobility of charge carriers.

15. Donor Atom Concentration

The donor concentration required depends on conductivity and mobility of electrons in the semiconductor.

16. Conductivity and Intrinsic

Resistivity

Conductivity depends on carrier concentration and mobility, while resistivity is the opposition offered to current flow.

17. Wavelength and Color of Radiation

When electrons recombine with holes, electromagnetic radiation is emitted. The wavelength determines the color of emitted light.

18. Voltage for Forward Current

In a p-n junction diode, a suitable forward voltage is required to allow significant current flow.

19. Formation of Ohmic and Schottky Junction
Ohmic Junction

A metal-semiconductor junction with very low resistance and current flow in both directions.

Schottky Junction

A metal-semiconductor junction showing rectifying behavior and fast switching.

20. Ohmic Contact

Ohmic contact is a low resistance metal-semiconductor contact that allows easy current flow in both directions.

21. Schottky Diode

A Schottky diode is a metal-semiconductor diode with low forward voltage drop and high switching speed.

22. Depletion Region

The depletion region is the area near a p-n junction where free charge carriers are absent due to recombination.

23. Forward and Reverse Biased p-n Junction

Forward Bias

The p-side is connected to the positive terminal and the n-side to the negative terminal, allowing current flow.

Reverse Bias

The p-side is connected to the negative terminal and the n-side to the positive terminal, allowing very little current flow.

24. Current-Voltage Characteristics of p-n Junction

In forward bias, current increases rapidly after cut-in voltage. In reverse bias, only a small leakage current flows until breakdown occurs.

25. Built-in Potential and Depletion

Width

Built-in potential is the internal voltage developed across a p-n junction, while depletion width is the width of the carrier-free region around the junction.

26. Factors Affecting Conductivity

Conductivity depends on:

Temperature
Carrier concentration
Mobility
Doping level
Impurity concentration
27. Cut-in Voltage and Breakdown Voltage
Cut-in Voltage

Minimum forward voltage required for significant current conduction.

Breakdown

Voltage

Reverse voltage at which reverse current increases suddenly.

28. DOS and Its Significance

Density of States (DOS) is the number of available energy states for electrons in a material. It helps determine carrier concentration and conductivity.

29. Temperature Effect on Intrinsic Carrier Concentration

As temperature increases, more electron-hole pairs are generated, increasing conductivity and decreasing resistivity.

30. n-type and p-type Semiconductor
n-type Semiconductor

Formed by adding pentavalent impurities. Electrons are majority